

HCI Notes: Cognitive Aspects

HCI - Week 4(a), Fall 2025

Abstract

These notes explore the fundamental cognitive aspects that underpin Human-Computer Interaction (HCI). The document begins by defining cognition and its importance in interaction design, introducing high-level concepts such as fast vs. slow thinking. It then provides a detailed examination of core cognitive processes, including attention, perception, memory, learning, and higher-order reasoning. For each process, we discuss its mechanisms, limitations, and direct design implications for creating effective and user-friendly interfaces. Finally, the notes cover major cognitive frameworks used in HCI to explain and predict user behavior, such as mental models, the gulfs of execution and evaluation, and more contemporary theories like distributed, external, and embodied cognition.

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1 Introduction to Cognition in HCI

The presentation begins by showing an image of PIA Flight PK 8303, which was involved in a tragic accident on May 22, 2020. This event, where human factors and cognitive load under pressure played a significant role, serves as a stark reminder of why understanding cognition is critically important in the design and operation of complex systems.

1.1 What is Cognition?

Cognition refers to the mental processes involved in gaining knowledge and comprehension. It is a broad term that encompasses a wide range of activities:

- Thinking, remembering, learning, daydreaming
- Decision-making, seeing, reading, talking, writing

At a higher level, cognition can be classified in several ways:

- **Experiential vs. Reflective Cognition (Norman, 1993):**
 - **Experiential Cognition** is a state of mind where people perceive, act, and react to events around them effectively and effortlessly. It is about being "in the moment" and relies on well-learned skills. Example: Driving a car on a familiar road.
 - **Reflective Cognition** involves thinking, comparing, and decision-making. It is the mode of cognition that leads to new ideas and creativity. Example: Planning a long road trip to an unfamiliar destination.
- **Fast vs. Slow Thinking (Kahneman, 2011):**
 - **Fast Thinking (System 1):** Operates automatically, intuitively, and without effort. It handles tasks like $2 + 2 = ?$ or identifying the color of your eyes.
 - **Slow Thinking (System 2):** Involves conscious attention and effortful mental activities. It is engaged for complex computations like $21 \times 29 = ?$ or tasks requiring deep thought, like recalling the name of your first school.

1.2 Why is Cognition Important for HCI?

Understanding cognition is fundamental to HCI for several reasons:

1. It provides knowledge about what users **can and cannot be expected to do**, setting realistic boundaries for design.
2. It helps to identify and explain the **nature and causes of problems** users encounter, allowing for more effective solutions.
3. It provides **theories, models, and methods** that lead to the design of better, more intuitive, and less error-prone interactive products.

2 Core Cognitive Processes

Several key cognitive processes are consistently relevant to interaction design.

2.1 Attention

- **Definition:** The process of selecting things on which to concentrate from the mass of stimuli around us. It allows us to focus on relevant information and filter out noise.
- **Types:** Attention can be *focused*, where we concentrate on one thing, or *divided*, where we attempt to attend to multiple things at once (multitasking).
- **Design Recommendation:** Information at the interface should be structured to capture users' attention when needed. This can be achieved using perceptual boundaries (windows), color, sound, or animation.

2.1.1 Attention and Information Layout

A study by Tullis (1987) demonstrated the importance of layout. Two screens with the same information density (31%) were tested.

- **Screen 1 (Cluttered):** Information was bunched together. Average search time: 5.5 seconds.
- **Screen 2 (Grouped):** Information was grouped into clear vertical categories using spacing. Average search time: 3.2 seconds.

Finding: Proper use of spacing and grouping (leveraging Gestalt principles) can significantly improve the efficiency of a visual search task by guiding the user's attention.

2.1.2 Multitasking and Attention

Is it possible to multitask without performance degrading? Research suggests it is very difficult.

- Multitasking can cause people to lose their train of thought, make errors, and have to start over.
- Studies (Ophir et al., 2009) have shown that "heavy multitaskers" are paradoxically *worse* at multitasking. They are more prone to distraction and find it harder to filter out irrelevant information.
- The primary reason multitasking is detrimental is that it **overloads our limited capacity to focus**.

2.1.3 Attention in Critical Contexts: Driving

Using a phone while driving is extremely dangerous because it involves divided attention.

- **Cognitive Load:** Drivers talking on a phone have longer reaction times because their cognitive resources are split between driving and the conversation. They often try to visualize the person they are talking to, which directly competes with the visual processing needed for the road.
- **Are hands-free phones safer?** No. The core problem is the cognitive distraction, not the physical act of holding a phone. A remote person on the phone is unaware of the driver's environment and will continue talking even when a hazard appears. In contrast, a passenger in the car shares the same context and will naturally pause the conversation, allowing the driver to refocus on the road.

Design Implications for Attention

- Make information salient (stand out) when it needs to be attended to. Use techniques like color, ordering, spacing, and animation.
- Avoid cluttering interfaces with too much information.
- Design to support effective task switching, helping users easily disengage and re-engage with an interface.

2.2 Perception

- **Definition:** The process of acquiring information from the world through our senses and transforming it into experiences.
- **Design Implication:** Representations must be readily perceivable. Text should be legible, and icons should be easy to distinguish and understand.

Effective visual design makes information easier to perceive. Studies have shown that grouping information using borders and adequate white space is more effective for quick scanning than using color contrast alone. Readability is also a key perceptual issue; text is easiest to read when there is high contrast between the text and the background (e.g., black on white).

Design Implications for Perception

- Icons should have clear and distinct meanings.
- Use bordering and spacing as effective ways to group related information.
- Ensure sounds are audible and distinguishable.
- Use proper color contrast. For example, yellow on white or green is a poor choice due to low contrast.
- Use haptic feedback judiciously, so it provides meaningful information without being overwhelming.

2.3 Memory

- **Definition:** The ability to encode, store, and recall knowledge and experiences.
- **Process:** We don't remember everything. Memory involves filtering what we attend to (**encoding**) and later retrieving it. The more attention paid to something and the more deeply it is processed, the more likely it is to be remembered.
- **Context:** The context in which a memory is encoded is crucial for retrieval. It can be difficult to recognize a person or recall information out of its usual context.

2.3.1 Recognition versus Recall

This is one of the most important concepts in HCI.

- **Recall:** Retrieving information from memory without a cue. Example: A command-line interface, where you must recall the exact command (e.g., 'ls -a').
- **Recognition:** Knowing you've seen something before when presented with it. Example: A graphical user interface (GUI) with menus and icons. You browse the options until you recognize the one you want.

Humans are much better at recognition than recall. This is why GUIs are generally easier to use than command-line interfaces.

2.3.2 The Misapplication of Miller's 7 ± 2 Rule

George Miller's (1956) famous finding that human short-term memory capacity is limited to roughly 7 ± 2 "chunks" of information has been widely misapplied in HCI. Designers sometimes insist that menus should have no more than seven items.

- **The Problem:** This rule applies to recalling arbitrary information from short-term memory. When using a GUI, users are **recognizing**, not recalling. They can easily scan a list of 20 menu items to find the one they want.
- **Conclusion:** The number of items on a menu should depend on the task and screen space, not an arbitrary number from a misapplied theory.

2.3.3 Personal Information Management (PIM)

We generate a huge amount of "digital stuff." The challenge is remembering what we named files and where we saved them.

- **Solution:** Design can facilitate this by providing rich metadata tools (e.g., time stamping, tagging, categories) and powerful search functions (e.g., Apple's Spotlight) that support recognition and contextual cues.

Design Implications for Memory

- Reduce cognitive load by avoiding complex procedures.
- **Design interfaces that promote recognition rather than recall.** This is a golden rule of HCI.
- Provide users with multiple ways to label and organize digital information (folders, categories, color-coding, etc.) to aid later retrieval.

2.4 Learning

- **Definition:** The accumulation of skills and knowledge.
- **Types:**
 - **Incidental Learning:** Effortless learning, like recognizing faces.
 - **Intentional Learning:** Effortful learning, like studying for an exam.
- **Key Insight:** People prefer to **learn by doing** and find it hard to learn by following instructions in a manual.

Design Implications for Learning

- Design interfaces that encourage exploration and are "safe" to explore.
- Constrain and guide learners to prevent them from getting lost or making critical errors.
- Use dynamic linking of concepts and representations to facilitate understanding of complex material.

3 Cognitive Frameworks

Cognitive frameworks are theories and models used to explain and predict user behavior at the interface.

3.1 Mental Models

- **Definition:** A user's internal understanding of how a system works. This knowledge is developed through learning and using the system.
- **Function:** People use their mental models to make inferences about how to carry out tasks and predict what the system will do next.
- **Erroneous Mental Models:** Often, users' mental models are incorrect or incomplete.
 - **Example: The Thermostat.** Many people believe that setting a thermostat to its highest temperature will make a room heat up faster. They are applying a "more is more" mental model (like a gas pedal). However, a thermostat is a simple on/off switch; it turns the heat fully on until the target temperature is reached and then turns it fully off. Setting it higher does not change the rate of heating.
- **Design Goal:** Design the user experience to help people build accurate mental models through clear instructions, tutorials, and a transparent and intuitive interface.

3.2 Gulfs of Execution and Evaluation

This framework by Don Norman describes the gaps between the user's goals and the system.

- **Gulf of Execution:** The gap between the user's goal and the actions required by the system. The user asks, "What can I do?" and "How do I do it?"

- **Gulf of Evaluation:** The gap between the system's state and the user's ability to perceive and understand it. The user asks, "What happened?" and "Is this what I wanted?"

Good design bridges these gulfs. It makes actions obvious (bridging execution) and provides clear, immediate feedback (bridging evaluation), reducing the user's cognitive effort.

3.3 Distributed Cognition

This framework views cognition not as an isolated process inside one person's head, but as a phenomenon **distributed across individuals, artifacts, and the environment**.

- It analyzes how information is propagated and transformed across different media (people's heads, computer displays, paper, verbal communication).
- **Example: An Airplane Cockpit.** The task of flying a plane is not performed by the pilot alone. It is distributed across the pilot, the co-pilot, air traffic control, and the various instruments and controls in the cockpit. Understanding this entire cognitive system is crucial for designing safe and effective interfaces.

3.4 External and Embodied Cognition

- **External Cognition:** Explains how we interact with external representations (maps, notes, diagrams) to augment our cognitive abilities.
 - **Cognitive Offloading:** Using external aids (like a to-do list or calendar) to reduce the effort of remembering.
 - **Computational Offloading:** Using tools and representations (like pen and paper for a long multiplication problem) to facilitate computation. The choice of representation matters immensely; multiplying 'CCXXXIII $\times$ CCCCXXXXVI' is nearly impossible compared to '234 $\times$ 456'.
- **Embodied Interaction:** Argues that cognition is shaped by our physical engagement with the world. We think *with* our bodies, not just our brains. Our active, physical experiences influence how we perceive, feel, and think.

4 Summary

- Cognition involves many processes, including attention, memory, perception, and learning.
- The design of an interface can greatly affect how well people can perform cognitive tasks.
- Theoretical frameworks like mental models and distributed cognition provide ways of understanding *why* and *how* people interact with products.
- By applying an understanding of cognition, designers can create products that are more intuitive, efficient, and enjoyable to use.